

# Virtual Cross Reference Matrix:

| Requirement ID | Requirement Description                                                                          | Verification Method | Verification Status | Evidence/Reference                                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------|---------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FVS-1          | Ingest Live Telemetry Data                                                                       | Unit Testing        | Completed           | <a href="https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/ingestion">https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/ingestion</a>           |
| FVS-2          | Utilizes Modern Game Engine                                                                      | Inspection          | Completed           | <a href="#">PDR Presentation</a>                                                                                                                                                                                                          |
| FVS-3          | Render a 3D air vehicle position and orientation update in real time                             | Demonstration       | Completed           | <a href="#">CDR Presentation</a>                                                                                                                                                                                                          |
| FVS-4          | Render and support for multiple air vehicle models                                               | Demonstration       | Completed           | <a href="#">CDR Presentation</a>                                                                                                                                                                                                          |
| FVS-5          | Maintain a smooth real-time experience, and must function at a minimum of 30 frames per second   | Performance Testing | Completed           | <a href="#">CDR Presentation</a>                                                                                                                                                                                                          |
| FVS-6          | Provide the ability to record and replay data from a data stream to allow for post-flight review | Unit Testing        | Completed           | <a href="https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/flightRecorder">https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/flightRecorder</a> |
| FVS-7          | Designed to run on the NVIDIA Jetson Orin Nano platform provided by the project sponsor          | Demonstration       | Completed           | <a href="#">CDR Presentation</a>                                                                                                                                                                                                          |
| FVS-8          | Support for multiple                                                                             | Unit Testing        | Completed           | <a href="https://github.com/S">https://github.com/S</a>                                                                                                                                                                                   |

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|        | camera views of the virtual environment (chase camera, fixed camera, and free camera)                                                                                                                     |                                |             | <a href="#">ME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/cameras</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| FVS-9  | Implement a GUI that allows users to start and stop data capture, select between air vehicle models, switch between playback modes, and control all configuration settings without modifying source code. | Unit Testing and Demonstration | Completed   | <a href="https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/GUI">https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/GUI</a><br><br><a href="https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/recordingPanel">https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/recordingPanel</a><br><br><a href="https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/playback">https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/playback</a> |
| FVS-10 | Support the real-time display of telemetry variables with display variables defined through a configuration file.                                                                                         | Unit Testing and Demonstration | In-progress |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| FVS-11 | The team shall provide documentation, including setup instructions, operating procedures, and a software bill of materials (SBOM)                                                                         | Inspection                     | Completed   | <a href="#">SBOM Document</a><br><br><a href="#">User Guide Document</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| FVS-12 | The application may use, but is not limited to, industry                                                                                                                                                  | Inspection                     | Completed   | <a href="#">PDR Presentation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|--------|----------------------------------------------------------------------------------------------|------------|-----------|-------------------------------------------------------------------------------------------------------------|
|        | standards such as C++, C#, and Python coding languages.                                      |            |           |                                                                                                             |
| FVS-13 | The project shall be designed using modular and open system methodologies.                   | Inspection | Completed | <a href="#">PDR Presentation</a>                                                                            |
| FVS-14 | The team shall use GitHub or GitLab to encourage configuration management and collaboration. | Inspection | Completed | <a href="https://github.com/SME-3D-Flight-Visualization">https://github.com/SME-3D-Flight-Visualization</a> |
| FVS-15 | The project shall be planned, designed, developed, and tested using Agile methodologies.     | Inspection | Completed | <a href="#">PDR Presentation</a>                                                                            |

## Known Bugs and Fixes:

| Bug Name:                             | Bug Description:                                                                                                                              | Patched Description: | Patched Progress: | Evidence/Reference |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------|--------------------|
| Invalid Config From sender            | Config from the sender is not the starting position of the drone                                                                              |                      | Incomplete        |                    |
| Flight Path Visualization in playback | While in playback, the flight path renders when going back to the old position, spikes and jumps instead of creating a smooth showcase of the |                      | Incomplete        |                    |

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